Tianhao Li

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Education Background

Duke University **August 2024 - May 2026 (Expected)**

Master of Science Candidate in Medical Physics

Research Focus: Safety & Privacy for Healthcare Foundation Model and Agentic System

Supervisors/Committee Member: Prof. Fang-Fang Yin (DKU), Prof. Zhenyu Yang (DKU), and Prof. Neil Gong (Duke)

North China University of Technology **September 2020 - June 2024**

Bachelor of Engineering in Information Security *GPA: 3.89/4 Grade: 90.88/100 transcript*

Core Courses: Applied Cryptography, Data Security, Image Processing, Principles of Computer Composition, Operating System, Reverse Engineering, Digital Forensic, Software Security, Network Attack and Defense

Honors: Outstanding Dissertation Award (Rank 1/101, Top 0.99%), Elite Student Scholarship (Rank 1/329, Top 0.30%)

Academia Research Experiences

University of Wisconsin, Madison **October 2024 – Present**

Research Intern @ Safe And secure Foundation mOdel systems Lab (SaFoLab) *Wisconsin, United States*

- **Duties**: Led a research team to advance the AgentFuzzer project, proposing a unified agent schema descriptor, AgentSchemaDescriptor. Developed the first systematic fuzz testing framework for LLM agents, addressing the challenge of improving generalizability in agent safety alignment testing.
- **Mentor**: Prof. Chaowei Xiao

Zhejiang University **July 2024 – October 2024**

Visiting Researcher @ Knowledge Engine Lab *Zhejiang, China*

- **Duties**: Led a 15-member research team on the SciSafeEval project, proposing the first safety benchmark for LLMs in scientific tasks. Identified misuse risks in SOTA LLMs (including GPT-4o, LLaMa3.2, QWen2.5) across biology, chemistry, medicine, and physics, and proposed mitigation strategies using Chain-of-Thought (CoT) and few-shot learning techniques.
- **Impacts**: arXiv:2410.03769 | Huggingface:Tianhao0x01/SciSafeEval | Github:DavidLee528/SciSafeEval
- **Mentors**: Prof. Qiang Zhang, Prof. Huajun Chen

High-Tech Innovation Institute **July 2022 – July 2023**

Research Assistant @ VR & AR Technology Innovation Center *Beijing, China*

- **Duties**: Led an 8-member R&D team developing a gait-based Intelligence, Surveillance, and Reconnaissance (ISR) system. Implemented a gait recognition SaaS prototype system and proposed security features including authentication, tracking, and early warning for physical attacks. Directed technical architecture design, technical reports, and research paper writing.
- **Impacts**: **Applied 100K+ CNY funding**, published 7 SCI/EI papers, won 4 national professional contests, delivered 3 oral presentations and 1 poster at international conferences.
- **Mentors**: Prof. Lijun Wang

Industry Research Experiences

TOPSEC *(with return offer)* **March 2024 – September 2024**

Security Researcher (Intern) @ Department of Innovative Technology Research *Beijing, China*

- **Duties**: **(i)** Contributed to the development of a pioneering prototype system for machine learning model security assessment, focusing on the design and implementation of a robust backend evaluation module. Successfully debugged and mitigated adversarial and poisoning attacks, enhancing the system's resilience and accuracy.; **(ii)** Contribute to a phishing email detection project, overseeing dataset collection and enhancement initiatives. Successfully applied Large Language Models (LLMs) and Supervised Fine Tuning (SFT) to develop a robust detection system.
- **Supervisor**: GM of Innovation Research Dr. Wei Wang

NSFOCUS **September 2023 – March 2024**

Security Researcher (Intern) @ Dubhe Lab, Innovation Institute *Beijing, China*

- **Duties**: **(i)** Design core algorithms and develop a prototype for NSFOCUS LLM Security Assessment System (LSAS); **(ii)** Enhance the security & privacy performance of NSFGPT and other private LLMs through offensive testing, i.e. red teaming; **(iii)** Conduct literature reviews encompassing research papers and Gartner's consulting reports on trustworthy AI, enriching the threat intelligence database; **(iv)** Write white paper on secure LLM.
- **Impacts**: The release of LSAS, SecLLM NSFGPT, and white paper Enhancing Network Security with SecLLM.
- **Supervisor**: Principal Security Researcher Dr. Xingkai Wang, CIO & GM of Innovation Research Dr. Wenmao Liu

Publications *(* for corresponding author, ^{1st} for co-first author)*

Safety Evaluation and Alignment of Generative Agent System

- Xiong, Q., Huang, Y., Jiang, Z., Chang, Z., Zheng, Y., **Li, T.**, & Li, M. (2025). Butterfly Effects in Toolchains: A Comprehensive Analysis of Failed Parameter Filling in LLM Tool-Agent Systems. arXiv preprint arXiv:2507.15296. (EMNLP 2025 Findings)

Safety Evaluation and Alignment of Generative Foundation Models

- **Li, T.**, Lu, J., Chu, C., Zeng, T., Zheng, Y., Li, M., Huang, H., Wu, B., Liu, Z., Ma, K., Yuan, X., Wang, X., Ding, K., Chen, H., & Zhang, Q.*. (2024). SciSafeEval: a comprehensive benchmark for safety alignment of large language models in scientific tasks. arXiv preprint arXiv:2410.03769. *(AAAI 2025 AICS Oral)*
- Zheng, Y.^{1st}, **Li, T.**^{1st}, Huang, H., Zeng, T., Lu, J., Chu, C., ... & Li, M. (2025). Are All Prompt Components Value-Neutral? Understanding the Heterogeneous Adversarial Robustness of Dissected Prompt in Large Language Models. arXiv preprint arXiv:2508.01554. *(AAAI 2026, submitted)*
- **Li, T.**, Zeng, T., Zheng, Y., Zhang, C., Lu, J., Huang, H., Chu, C., Yin, F.F. & Yang, Z. (2025). Synthetic Poisoning Attacks: The Impact of Poisoned MRI Image on U-Net Brain Tumor Segmentation. arXiv preprint arXiv:2502.03825. *(ICLR 2025 SynthData, AAPM 2025 Oral)*
- Zhang, Q., Ding, K., Li, M., **Li, T.**, Liu, D., & Chen, H. (2025). Comprehensive Alignment for BioMolecular Large Language Models. *Nature Computational Science (JCR-Q1, Peer Review)*
- Ghosh, S., ..., **Li, T.**, ..., Mattson, P., Liang, P., Vanschoren, J.. (2025). AILuminate: Introducing v1.0 of the AI Risk and Reliability Benchmark from MLCommons. arXiv preprint arXiv:2503.05731.

Learning-Based Pattern Recognition and Signal Understanding

- **Li, T**., Ma, W., Zheng, Y., Fan, X., Yang, G., Li, Y., Wang, L., & Li, Z.*. (2024). A Survey on Gait Recognition Against Occlusion: Taxonomy, Dataset and Methodology. *PeerJ Computer Science (**JCR-Q1**)*
- Wei, Y., Liu, X., Xu, C., Zhang, G., **Li, T.**, Yuan, W., Ho, H. P., & Xu, M. (2025). From Droplets to Diagnosis: AI-Driven Imaging and System Integration in Digital Nucleic Acid Amplification Testing. *Biosensors and Bioelectronics (**JCR-Q1**, Accepted)*
- Ma, W., Li, Y.*, **Li, T.**, Yang, H., Li, Z., Wang, L., & Xuan, J.. (2024). SFSWTS: EEG Emotion Recognition Based on Spatial-frequency Shifted Windows Time Self-attention Neural Network. *Neurocomputing (**JCR-Q1**)* [PDF]
- Ma, W., Zheng, Y., **Li, T.**, Li, Z., Li, Y., & Wang, L.*. (2024). A comprehensive review of deep learning in EEG-based emotion recognition: classifications, trends, and practical implications. *PeerJ Computer Science (**JCR-Q1**)* [PDF]
- **Li, T.**, Zheng, Y., Zhang, H., Ma, W., & Li, Y.*. (2023). Research on Real-time Early Warning Mechanism of Aviation Safety Based on Finite State Machine Underlying in QAR Stream Data. *Proceedings of the 2023 7th International Conference on Big Data and Internet of Things (BDIoT)* [PDF][Oral][Slide]

Honors and Awards

- Aug.2024, Second Prize in 17th CISCN Security Project Contest [certificate] *National*
- Jul.2024, Top 5 out of 89 in Galaxy Generative AI Safety Contest Attack Track *National*
- Feb.2024, S Prize in 2024 Mathematical Contest In Modeling (MCM) [paper] *National*
- Nov.2023, Second Prize in 11th Digital Media Technology and Creativity Contest [certificate] *National*
- Nov.2023, Third Prize in 8th National Cryptography Contest [certificate][list][news] *National*
- Aug.2023, Third Prize in 16th CISCN Security Project Contest [cert.][report][slide][video] *National*
- Jun.2023, Third Prize in 16th CISCN AWDP (Attack with Defense Plus) Contest [certificate] *Regional*
- Apr.2023, First Prize in 13th MathorCup Mathematical Modeling Challenge [cert.][paper] *National*
- Apr.2023, First Prize in 14th Lanqiao Software Development and Algorithm Contest [certificate] *Provincial*
- Jun.2022, Winning Prize in 15th CISCN Security Project Contest [certificate][report][slide] *National*
- Jun.2022, Third Prize in 15th CISCN CTF (Capture the Flag) Contest [certificate] *Regional*
- Jun.2021, Third prize in 12th Lanqiao Software Development and Algorithm Contest [cert.] *National*
- Apr.2021, First Prize in 12th Lanqiao Software Development and Algorithm Contest [certificate] *Provincial*

Open Source Activities

- NVIDIA/garak (Generative AI Red-teaming Tool Kit. Metasploit in the field of LLM Security. ∼**4.7K Stars**)

  (i) Robustness Enhancement: Python interpreter version bug fix#296; Replicate LLM generator exception error fix#401; Add exception handling logic to the Detector base class#566; (ii) New Feature: Multi-modal Jailbreaking Attack on LLaVA #587 (Make it the **world's first** tool with multimodal red teaming capabilities.); Prompt Architecture Enhancement for Better Multi-modal Red Teaming (#658); Introducing new probe SciSafeEval (#948); Network Proxy Feature For Generator (garak:proxy_dev) (v0.9.0.9, v0.9.0.11, v0.9.0.13)

Academic Talks

- Invited talk - Toward Trustworthy Generative Foundation Models and Autonomous Agent System (25 mins) @ Institute of Software, Chinese Academy of Sciences, Aug 2025, Beijing, China [Link]
- Invited talk - Guardians of Trust: Safety & Privacy for Healthcare Foundation Models (15 mins) @ Duke Kunshan Graduate Student Colloquium Apr 2025, Jiangsu, China [Link]
- Contributed talk - SciSafeEval: A Comprehensive Benchmark for Safety Alignment of Large Language Models in Scientific Tasks (20 mins) @ The 39th Annual AAAI Conference on Artificial Intelligence (AAAI-2025) Artificial Intelligence for Cyber Security (AICS), Mar 2025, Philadelphia, PA, USA [Link]

Professional Services

Journal Referee

- ACM Transactions on Intelligent Systems and Technology (1);
- IEEE Transactions on Artificial Intelligence (2);
- IEEE Transactions on Biomedical Engineering (2);
- Expert Systems With Applications (1);
- Engineering Applications of Artificial Intelligence (4);
- Computer Networks (2);
- Journal of Biomedical and Health Informatics (3);
- International Journal of Human-Computer Interaction (1);
- Reliability Engineering & System Safety (2);

Program Committee Member

- AAAI Conference on Artificial Intelligence (AAAI) 2025(1), 2026(7);
- Association for Computational Linguistics Rolling Review (ACL ARR) Feb(2), May(4) 2025;
- International Joint Conference on Artificial Intelligence (IJCAI) 2025(3);
- International Conference on Computational Linguistics (COLING) 2025(6);
- International Joint Conference on Neural Networks (IJCNN) 2025(3);
- IEEE International Symposium on Biomedical Imaging (ISBI) 2025(1);
- Conference on Health, Inference, and Learning (CHIL) 2025(2);
- Workshop on LLM Security (LLMSEC, with ACL) 2025(1);
- Workshop on Privacy-Preserving Artificial Intelligence (PPAI, with AAAI) 2025(3);
- Workshop of Deployable AI (DAI, with AAAI) 2025(3);
- Multi-Agent AI in the Real World (MARW, with AAAI) 2025(2);
- Workshop on Preventing and Detecting LLM Misinformation (PDLM, with AAAI) 2025(3);
- International Workshop on AI Governance (AIGOV, with AAAI) 2025(1);
- Workshop on Foundation Models in the Wild (FM-Wild, with ICLR) 2025(2);
- Workshop on Deep Generative Model in Machine Learning (DeLTa, with ICLR) 2025(2);
- From Understanding Model Behavior to Discovering New Scientific Knowledge (XAI4Science, with ICLR) 2025(1);
- Workshop on GenAI Watermarking (WMARK, with ICLR) 2025(1);
- Workshop on AI for Children: Healthcare, Psychology, Education (AI4CHL, with ICLR) 2025(2);

Membership

- MLCommon/AI Risk & Reliability (AIRR) Working Group — Member/Individual Contributor
- Institute of Electrical and Electronics Engineers (IEEE) — Graduate Student Member
- Association for Computing Machinery (ACM) — Graduate Student Member
- Association for the Advancement of Artificial Intelligence (AAAI) — Graduate Student Member
- China Computer Federation (CCF) — Graduate Student Member

Leadership

- Special Interest Group in Modern Interdisciplinary Research ⇒ sigmir.org — Founder & Head of Research

  In January 2025, I founded SIGMIR, a global research group primarily composed of graduate students. Beyond publishing academic papers, SIGMIR drives industrial applications, consistently delivering high-quality research and technological advancements to both academia and industry. (Non-profit under section 501(c)(3))

Media Coverage

- AI Safety in China #17 [Article] — Substack
- AI Action Summit: Public Interest AI [PDF] — Concordia AI
- AI Action Summit: AI of Trust - Security & Safety [PDF] — Concordia AI
- Student Voices | "Transfer Learning": From CS to Safer and More Reliable Medical AI [Article] — DKU Med. Phy.